



ZOMATO

DATA ANALYSIS

2025



DATA ANALYSIS FILE LINK:
[Zomato Colab File](#)

PRESENTED BY :
APOORVA

+91 9560569849
Apoorvasharma1971@gmail.com

Table Of Content

03 ABOUT THE COMPANY

04 AIM

05 OBJECTIVE

06 TOOLS AND FORMULAS

07 DATA OVERVIEW

08 BASIC STATISTICS

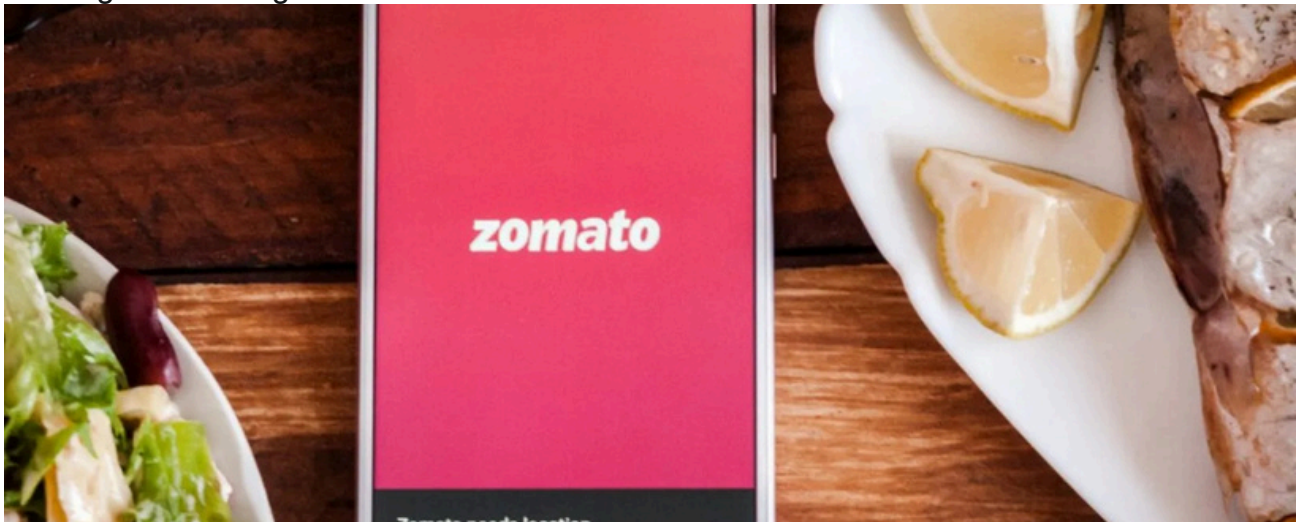
**09 GRAPHS, INSIGHTS
AND ANALYSIS**

**10 CONCLUSION AND
ACKNOWLEDGEMENT**



ABOUT THE COMPANY

Zomato is one of India's leading online food delivery and restaurant aggregator platforms, founded in 2008 by Deepinder Goyal and Pankaj Chaddah. What began as a simple restaurant discovery platform has now transformed into a global company providing food delivery, dining-out services, restaurant reviews, online ordering, and even hyperlocal grocery delivery through various integrations. Headquartered in Gurugram, Zomato has expanded its operations across multiple countries, becoming a household name for millions of users who rely on it daily to browse menus, explore cuisines, and get food delivered conveniently at their doorstep. Its user-friendly mobile application and website have significantly reshaped the food service industry by connecting restaurants with consumers on a large scale. With its business model based on data-driven insights, customer preferences, and seamless digital transactions, Zomato has positioned itself as a tech-first enterprise that thrives on consumer behavior analysis, market demand patterns, and real-time logistics management.



From a data analysis perspective, Zomato presents an extremely rich and dynamic case study due to the vast amount of data it generates on a daily basis. The platform collects and processes information related to user preferences, restaurant ratings, delivery timelines, peak order hours, geographic demand, and customer feedback. These datasets provide valuable opportunities for analyzing consumer trends, predicting demand fluctuations, optimizing delivery networks, and enhancing overall customer satisfaction. Zomato also leverages advanced algorithms and machine learning models to personalize recommendations, detect anomalies such as fake reviews, and improve operational efficiency for partner restaurants. By continuously using data to refine its services, the company not only enhances user engagement but also creates a strong competitive advantage in the highly competitive food delivery industry. Therefore, analyzing Zomato's data ecosystem is vital to understanding how modern tech-based platforms innovate, sustain growth, and adapt to evolving customer expectations in the digital economy.

AIM

About This Project :

In this project, we aim to analyze Zomato restaurant data to identify key factors that contribute to the success of restaurants, as measured by their ratings. By exploring various features such as location, cuisine, pricing, and service offerings, we aim to provide insights that can help restaurant owners and Zomato users make informed decisions.

A circular diagram representing the first step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Business Understanding" is centered. Below the circle, there is a small icon of a bar chart with an upward-pointing arrow.

Business Understanding

STEP 1

A circular diagram representing the second step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Data Understanding" is centered. Below the circle, there is a small icon of a database cylinder.

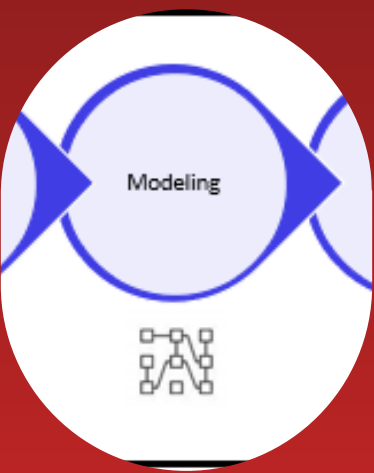
Data Understanding

STEP 2

A circular diagram representing the third step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Data Preparation" is centered. Below the circle, there is a small icon of a data table with a filter arrow.

Data Preparation

STEP 3

A circular diagram representing the fourth step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Modeling" is centered. Below the circle, there is a small icon of a neural network diagram.

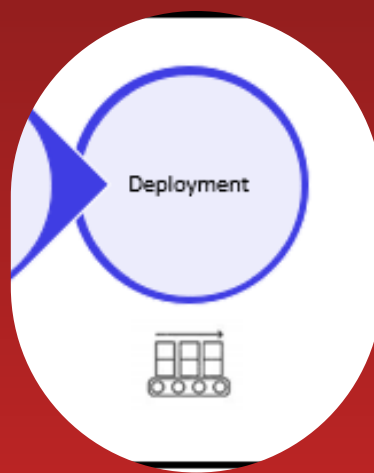
Modeling

STEP 4

A circular diagram representing the fifth step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Evaluation" is centered. Below the circle, there is a small icon of a checklist or document with a checkmark.

Evaluation

STEP 5

A circular diagram representing the sixth step of the project. It features a light blue circle with a darker blue border. Inside the circle, the text "Deployment" is centered. Below the circle, there is a small icon of a server rack or data center.

Deployment

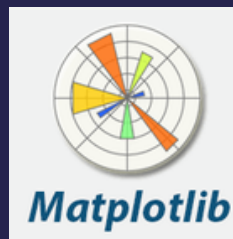
STEP 6

OBJECTIVE

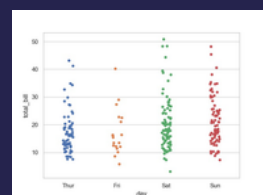
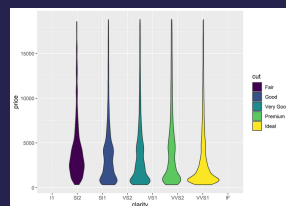
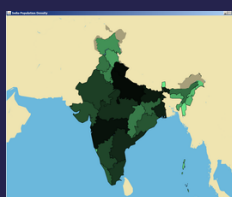
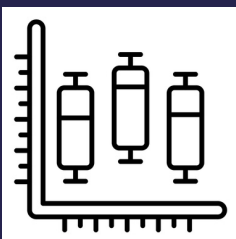
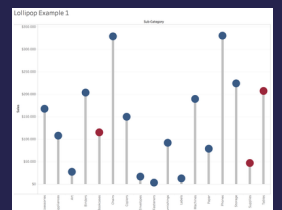


- Knowing basic composition of data
- Removing duplicates
- Dealing with missing values
- Understanding features
- Plotting horizontal bar charts (multicolor)
- Using groupby, apply, and unique functions
- Scatter plot
- Word Cloud Box plot
- Density plot
- Bar Charts
- Drawing insights and conclusions from data

TOOLS



GRAPHS



DATA OVERVIEW

```
[4] pd.set_option('display.max_columns', None)
```

```
df.columns.unique()
```

```
Index(['name', 'establishment', 'city', 'city_id', 'locality', 'latitude',
       'longitude', 'country_id', 'locality_verbose', 'cuisines', 'timings',
       'average_cost_for_two', 'price_range', 'highlights', 'aggregate_rating',
       'rating_text', 'votes', 'opentable_support', 'delivery',
       'Online_Available'],
      dtype='object')
```

```
df.nunique()
```

	0
res_id	55568
name	41100
establishment	27
url	55568
address	50657
city	99
city_id	83
locality	3731
latitude	53362
longitude	53326
zipcode	1311
country_id	1
locality_verbose	3910
cuisines	9382
timings	7740
average_cost_for_two	145
price_range	4
currency	1
highlights	31455
aggregate_rating	33
rating_text	39

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211944 entries, 0 to 211943
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   res_id                211944 non-null  int64
1   name                  211944 non-null  object
2   establishment         211944 non-null  object
3   url                   211944 non-null  object
4   address               211810 non-null  object
5   city                  211944 non-null  object
6   city_id               211944 non-null  int64
7   locality              211944 non-null  object
8   latitude              211944 non-null  float64
9   longitude             211944 non-null  float64
10  zipcode               48757 non-null   object
11  country_id            211944 non-null  int64
12  locality_verbose      211944 non-null  object
13  cuisines              210553 non-null  object
14  timings               208070 non-null  object
15  average_cost_for_two  211944 non-null  int64
16  price_range           211944 non-null  int64
17  currency              211944 non-null  object
18  highlights            211944 non-null  object
19  aggregate_rating      211944 non-null  float64
20  rating_text           211944 non-null  object
21  votes                 211944 non-null  int64
22  photo_count           211944 non-null  int64
23  opentable_support     211896 non-null  float64
24  delivery              211944 non-null  int64
25  takeaway              211944 non-null  int64
dtypes: float64(4), int64(9), object(13)
memory usage: 42.0+ MB
```

```
df.isnull().sum()
```

	0
res_id	0
name	0
establishment	0
url	0
address	134
city	0
city_id	0
locality	0
latitude	0
longitude	0
zipcode	163187
country_id	0
locality_verbose	0
cuisines	1391
timings	3874
average_cost_for_two	0
price_range	0
currency	0
highlights	0
aggregate_rating	0
rating_text	0

BASIC STATISTICS

```
[ ] df.describe()
```

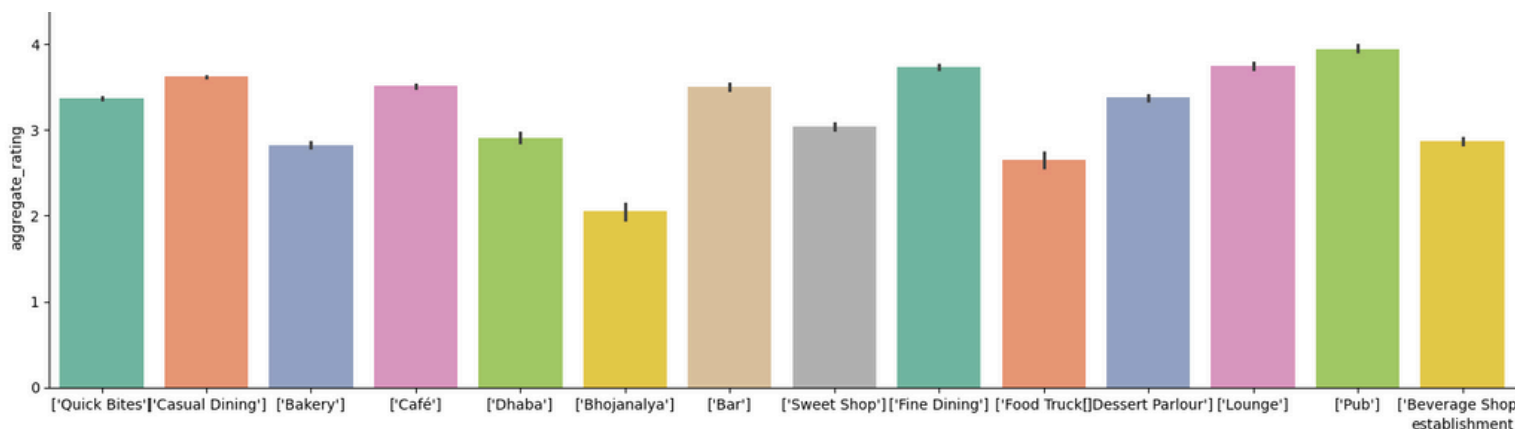


	city_id	latitude	longitude	average_cost_for_two	price_range	aggregate_rating	votes	opentable_support	delivery
count	211944.000000	211944.000000	211944.000000	211944.000000	211944.000000	211944.000000	211944.000000	211896.0	211944.000000
mean	4746.785434	21.499758	77.615276	595.812229	1.882535	3.395937	378.001864	0.0	-0.255907
std	5568.766386	22.781331	7.500104	606.239363	0.892989	1.283642	925.333370	0.0	0.964172
min	1.000000	0.000000	0.000000	0.000000	1.000000	0.000000	-18.000000	0.0	-1.000000
25%	11.000000	15.496071	74.877961	250.000000	1.000000	3.300000	16.000000	0.0	-1.000000
50%	34.000000	22.514494	77.425971	400.000000	2.000000	3.800000	100.000000	0.0	-1.000000
75%	11306.000000	26.841667	80.219323	700.000000	2.000000	4.100000	362.000000	0.0	1.000000
max	11354.000000	10000.000000	91.832769	30000.000000	4.000000	4.900000	42539.000000	0.0	1.000000

The `df.describe()` output summarizes the dataset of 211,944 records, offering insights into restaurant attributes. The `city_id` column ranges from 1 to 11,354, representing unique cities, though its mean and standard deviation are less meaningful since IDs are categorical. The latitude and longitude values mostly align with India's geography (median ~22.5, 77.4), but extreme outliers like latitude 10,000 or longitude 0 indicate errors requiring cleaning. The `average_cost_for_two` shows wide variation: mean ~596, median 400, and maximum 30,000, reflecting both budget and luxury options, with some invalid zeros. The `price_range` (1–4) clusters around 2, indicating most restaurants are mid-priced. Customer sentiment is reflected in `aggregate_rating`, ranging 0–4.9 with a mean of 3.39 and median of 3.8, suggesting generally positive reviews, though zeros may mean unrated entries. The `votes` column is skewed: median 100 but maximum over 42,000, showing a few highly popular restaurants dominate engagement. `opentable_support` is almost entirely zero, meaning reservations via OpenTable are rare, while `delivery` values (–1 to 1) reveal inconsistent coding of service availability. Overall, the summary highlights a large, diverse dataset with mid-range affordability and favorable ratings but also clear anomalies and missing data that require preprocessing before deeper analysis.

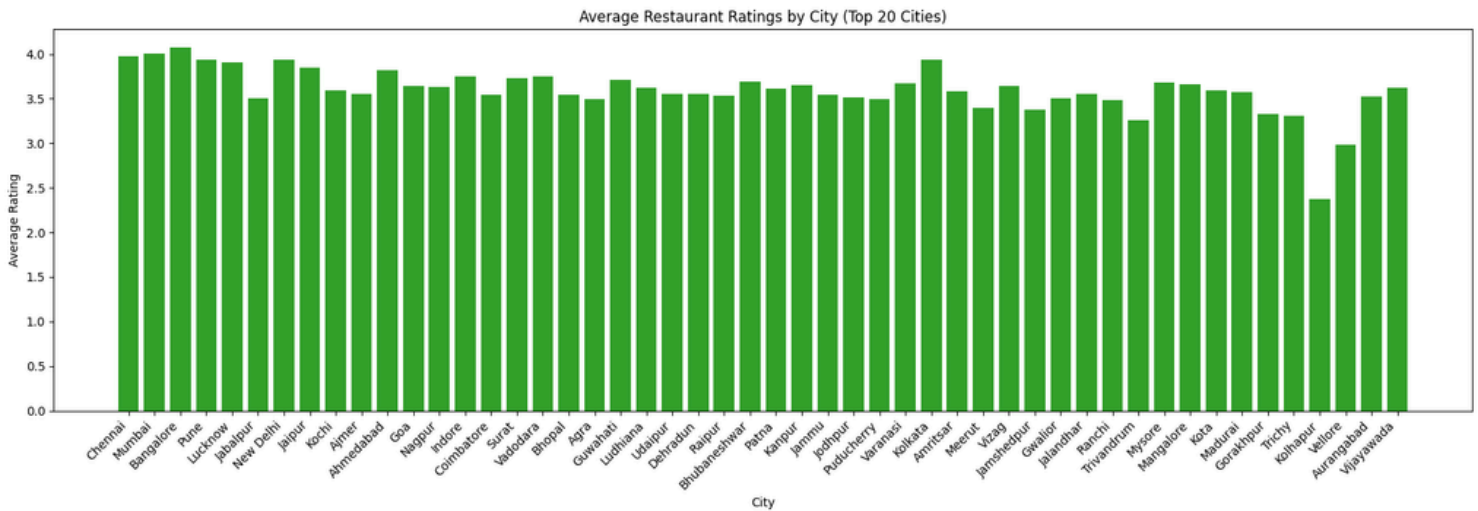
GRAPHS, INSIGHTS AND ANALYSIS

This bar chart represents the average aggregate ratings of different types of restaurant establishments found in the Zomato dataset. Each bar corresponds to a particular establishment type such as Quick Bites, Casual Dining, Bakery, Café, Dhaba, Fine Dining, Lounge, Pub, Microbrewery, Cocktail Bar, and others. The y-axis indicates the average aggregate rating, which ranges roughly between 1.0 and 4.5. Establishments like Microbrewery, Cocktail Bar, and Pub appear to have some of the highest average ratings, while categories such as Paan Shop, Mess, and Confectionery are rated significantly lower. This visualization highlights clear differences in customer preferences and satisfaction levels across various dining formats.



Average Customer Ratings Across Different Restaurant Establishments

From the chart, it is evident that premium or experiential dining formats such as Fine Dining, Lounge, Pub, and Microbrewery consistently attract higher ratings, suggesting that customers value ambiance and overall dining experience in addition to food quality. On the other hand, quick-service outlets like Paan Shop, Kiosk, and Mess tend to score lower, reflecting their functional rather than experiential nature. These insights can help businesses and stakeholders in the food industry better understand customer expectations, enabling them to design strategies for improving service quality, enhancing customer experience, and positioning their establishments more effectively in the market.

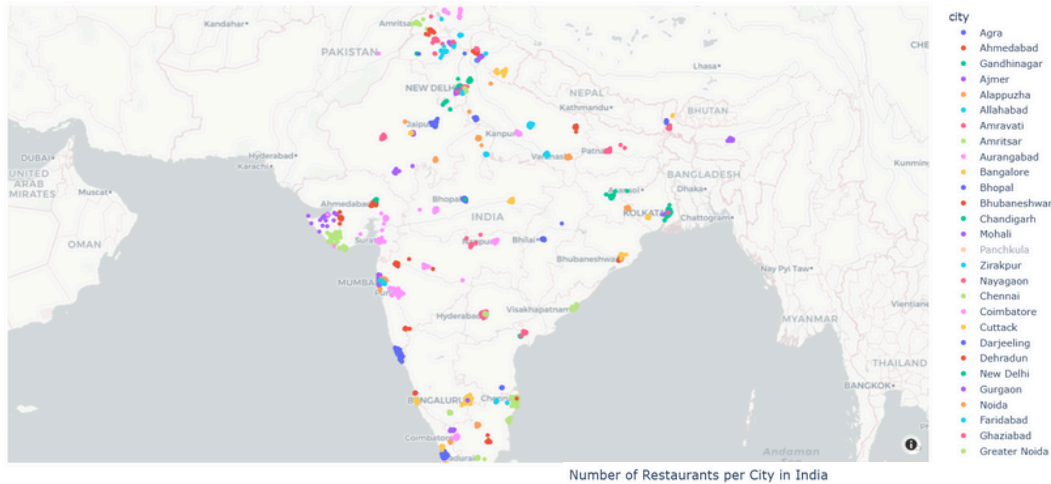


Average Restaurant Ratings by City (Top 20 Indian Cities)

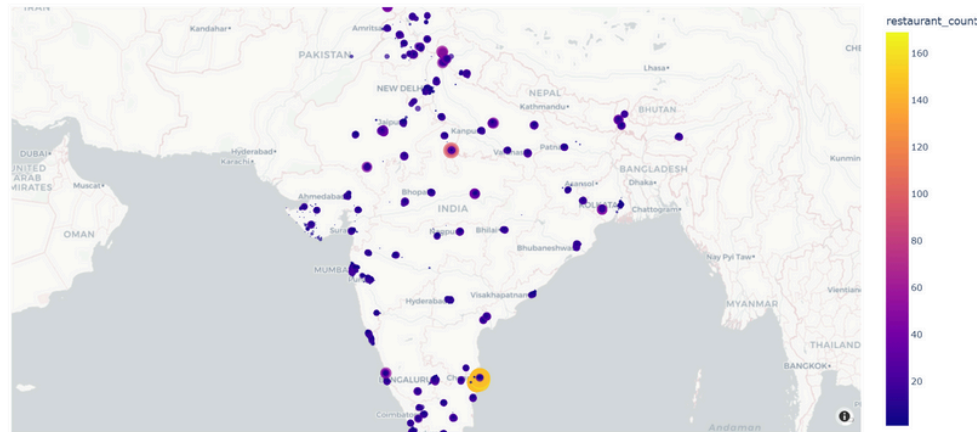
This bar chart illustrates the average restaurant ratings across the top 20 Indian cities in the Zomato dataset. The y-axis shows the average rating on a scale of 0 to 5, while the x-axis lists cities such as Chennai, Mumbai, Bangalore, Pune, Lucknow, Delhi, Jaipur, Ahmedabad, Kolkata, Amritsar, and more. Most cities report ratings in the range of 3.5 to 4.0, indicating a generally positive customer experience across major urban centers. Cities like Pune, Bangalore, and Amritsar stand out with higher average ratings, exceeding 4.0 in some cases, reflecting a strong dining culture and customer satisfaction. On the other hand, cities such as Aurangabad and Vellore record significantly lower averages, suggesting scope for improvement in restaurant quality and services.

The visualization highlights regional variations in customer perceptions of dining experiences. Metropolitan cities such as Mumbai, Delhi, Bangalore, and Chennai consistently show above-average ratings, likely due to better infrastructure, more competitive restaurant markets, and diverse culinary options. Smaller cities, however, display more fluctuation, with some like Amritsar and Vijayawada performing surprisingly well, while others such as Aurangabad lag behind. These insights are valuable for both customers and businesses: customers can identify cities with the best-rated dining experiences, while restaurant owners and policymakers in underperforming areas can focus on improving quality, service, and variety to boost customer satisfaction.

Restaurants in India (each dot = one restaurant)

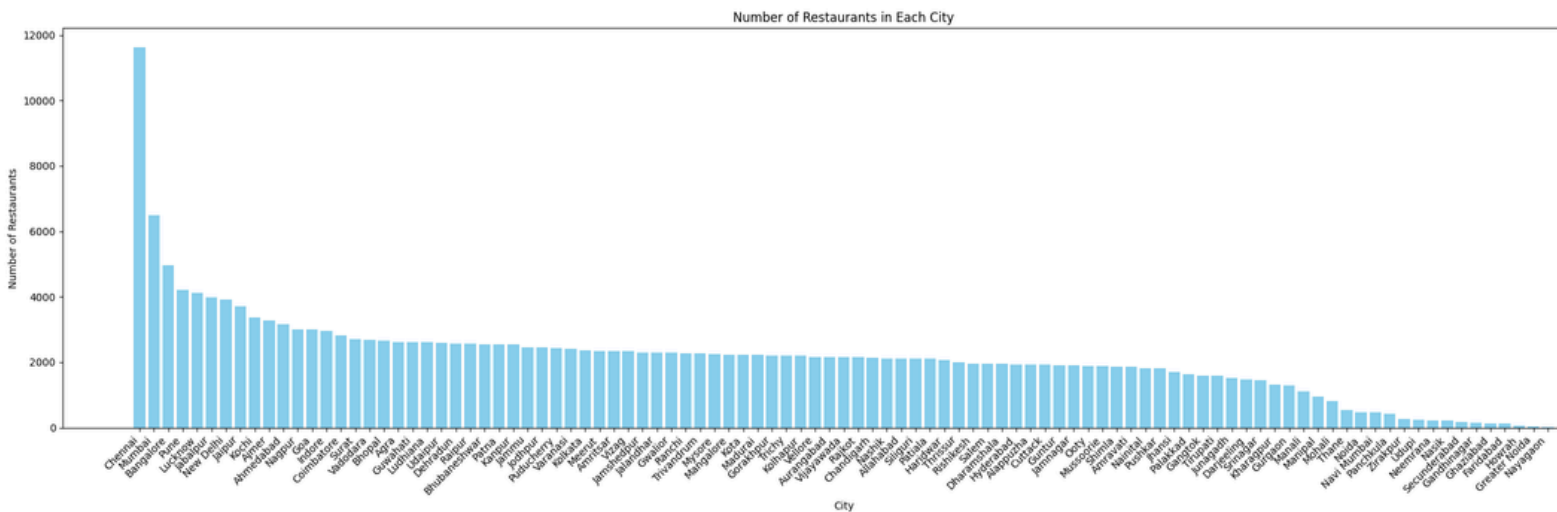


Number of Restaurants per City in India



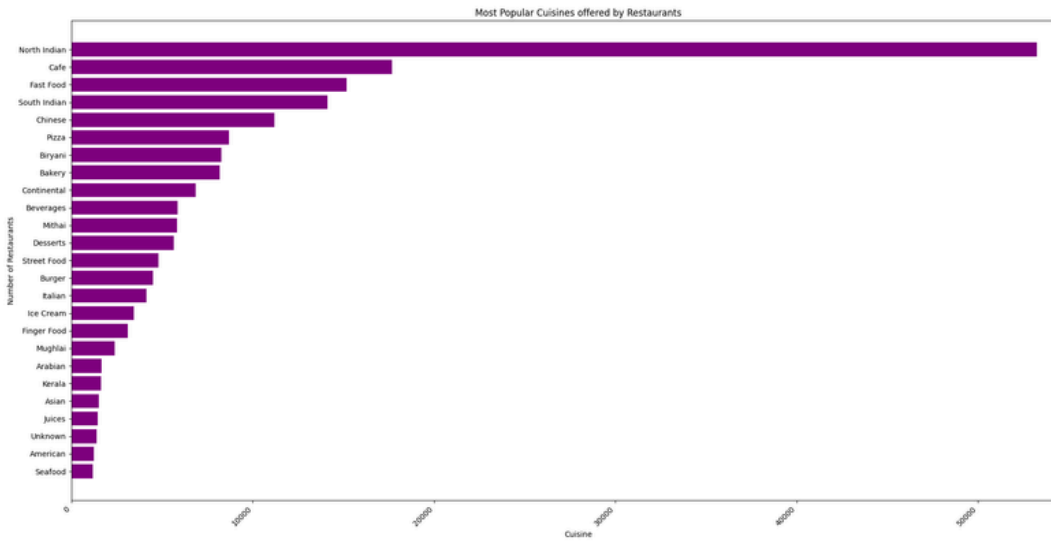
Geographic Distribution and Density of Restaurants in India

The first map visualizes the geographic distribution of restaurants across India, where each colored dot represents a restaurant and the color corresponds to the respective city. This gives a clear idea of how restaurants are spread across different regions, with noticeable clusters in metropolitan areas like Delhi, Mumbai, Bangalore, Chennai, and Kolkata. The second map provides a density-based view by showing the number of restaurants per city, represented by varying sizes and colors of circles. Larger and brighter circles (towards yellow) indicate cities with a higher concentration of restaurants, such as Chennai, Bangalore, and New Delhi, while smaller dark circles show cities with fewer restaurants. Together, these maps highlight both the widespread presence of Zomato-listed restaurants across India and the dominance of certain urban centers as major hubs for food services.



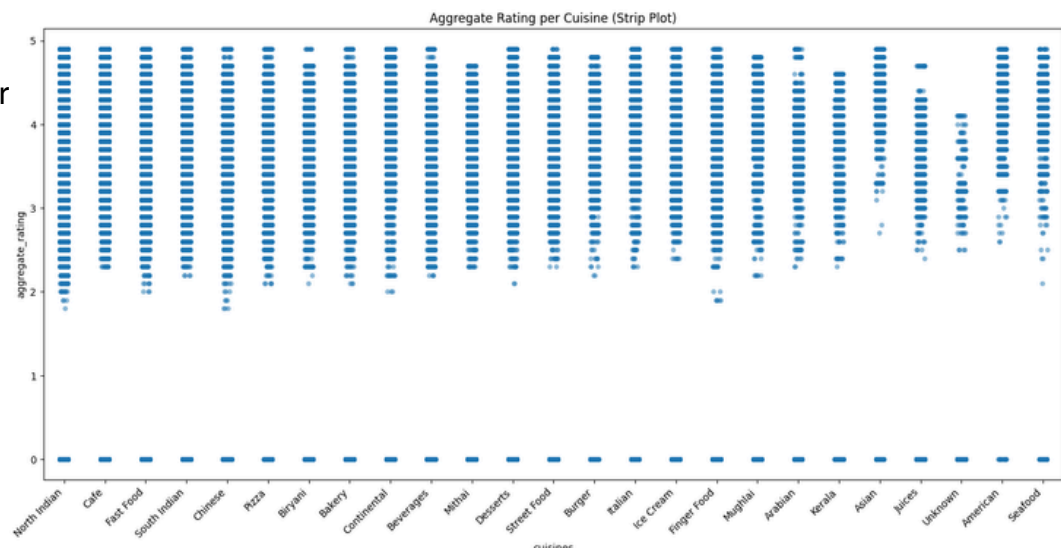
Number of Restaurants Across Indian Cities

The bar chart illustrates the distribution of restaurants across various Indian cities, highlighting the dominance of a few urban centers. Chennai leads with the highest number of restaurants, followed by Mumbai, Bangalore, and Pune, which together form the major hubs of food services in India. Other cities like Lucknow, Kolkata, New Delhi, and Jaipur also show significant restaurant presence, though comparatively smaller in scale. Beyond the top metros, the number of restaurants gradually declines across tier-2 and tier-3 cities, reflecting the concentration of Zomato-listed outlets in metropolitan and highly populated regions. This visualization clearly shows the uneven distribution, where a few cities contribute disproportionately to the overall restaurant count, while smaller towns and suburban areas remain underrepresented.



Most Popular Cuisines Offered by Restaurants

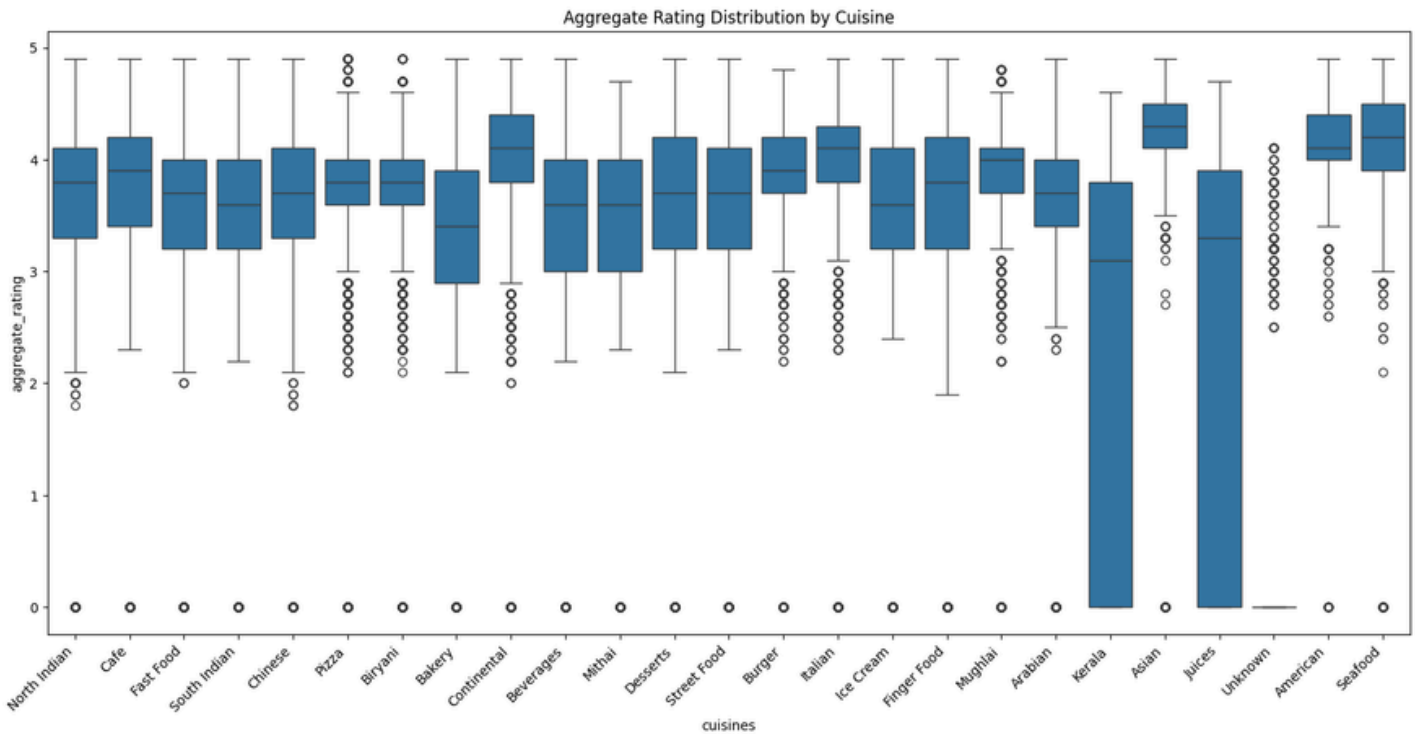
The bar chart showcases the popularity of different cuisines offered by restaurants across India. North Indian cuisine dominates overwhelmingly, followed by Cafes, Fast Food, and South Indian cuisine, reflecting the strong cultural and regional food preferences of the population. Chinese and Pizza also appear as widely available options, highlighting the influence of global and fusion cuisines in urban food markets. Other popular categories include Biryani, Bakery, Continental, Beverages, and Mithai, which indicate a balance between traditional Indian flavors and modern dining trends. Meanwhile, cuisines such as Mughlai, Arabian, Asian, and Seafood are comparatively less common, catering to more niche or region-specific audiences. Overall, the chart underlines the diversity of India's food industry, where traditional Indian cuisines dominate but international and contemporary offerings are steadily gaining ground.



Aggregate Ratings Across Different Cuisines

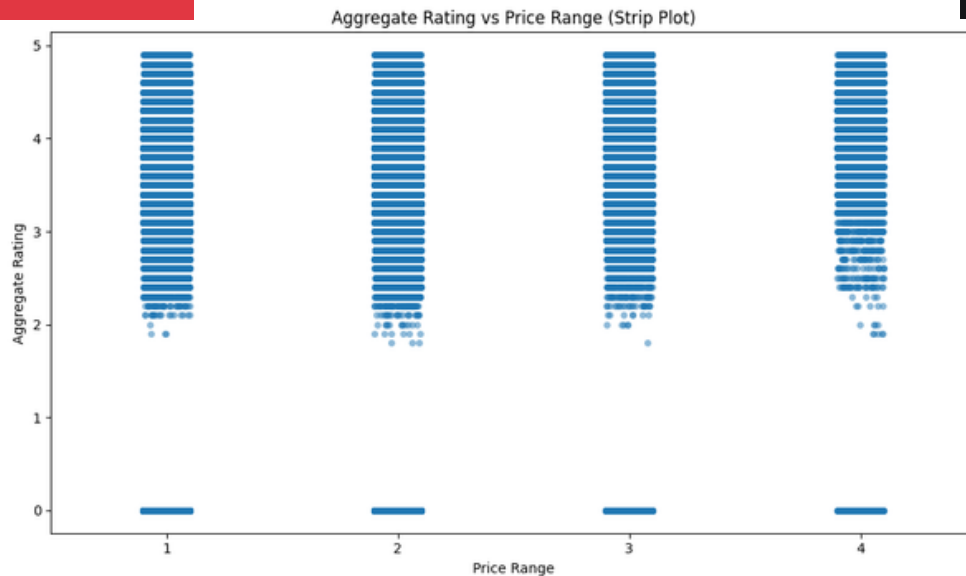
The strip plot displays the distribution of aggregate ratings for restaurants based on different cuisines. Most cuisines, including North Indian, South Indian, Chinese, and Fast Food, show a wide spread of ratings between 2 and 5, with a majority of restaurants concentrated around ratings above 3.5, indicating generally positive customer feedback. Cuisines like Continental, Italian, and Asian often achieve higher ratings, suggesting strong customer

satisfaction, while Street Food and regional cuisines exhibit slightly more variability. A notable observation is that almost all cuisines have a few restaurants with very low ratings (close to 0), which could be outliers or poorly performing outlets. Overall, the visualization emphasizes that while North Indian and South Indian cuisines dominate in terms of presence, customer satisfaction trends remain fairly consistent across both Indian and international cuisines, with most ratings clustering in the favorable range



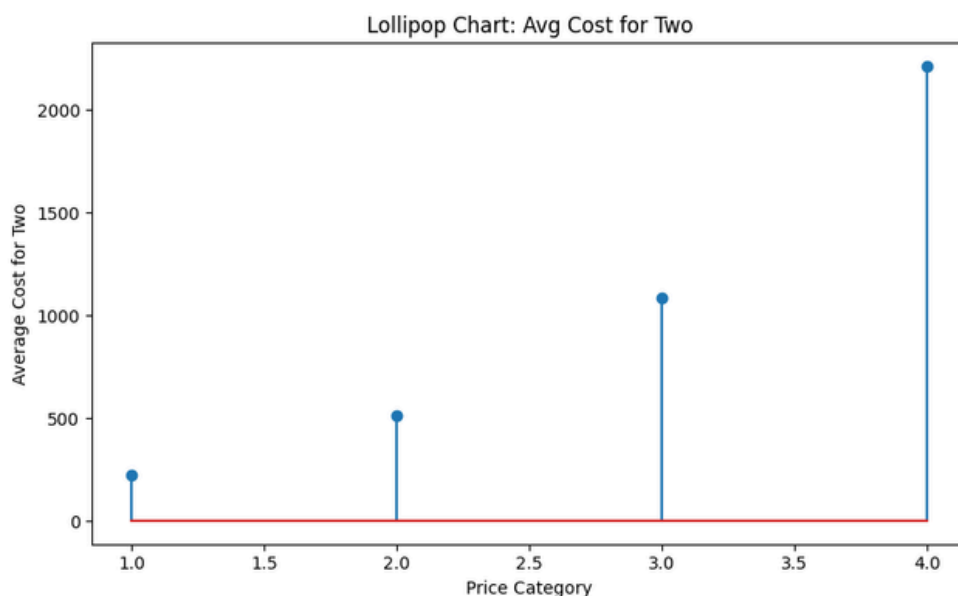
Aggregate Rating Distribution by Cuisine

The boxplot illustrates how customer ratings vary across different cuisines. Most cuisines—including North Indian, South Indian, Fast Food, Chinese, and Biryani—have ratings concentrated between 3.0 and 4.5, suggesting overall favorable customer satisfaction. International cuisines such as Italian, Continental, Asian, and Seafood display slightly higher medians (around 4.0), highlighting consistent positive feedback. On the other hand, regional cuisines like Kerala and “Unknown” categories show greater variability, with some extreme outliers and a few ratings near zero, which likely represent either niche establishments or underperforming outlets. Notably, cuisines like Juices, Ice Cream, and Desserts tend to achieve higher consistency in ratings, reflecting strong customer appreciation in those categories. Overall, the plot indicates that while popular cuisines dominate in numbers, customer satisfaction remains fairly balanced across both Indian and international food segments, with most ratings leaning toward the positive end.



Aggregate Rating vs Price Range

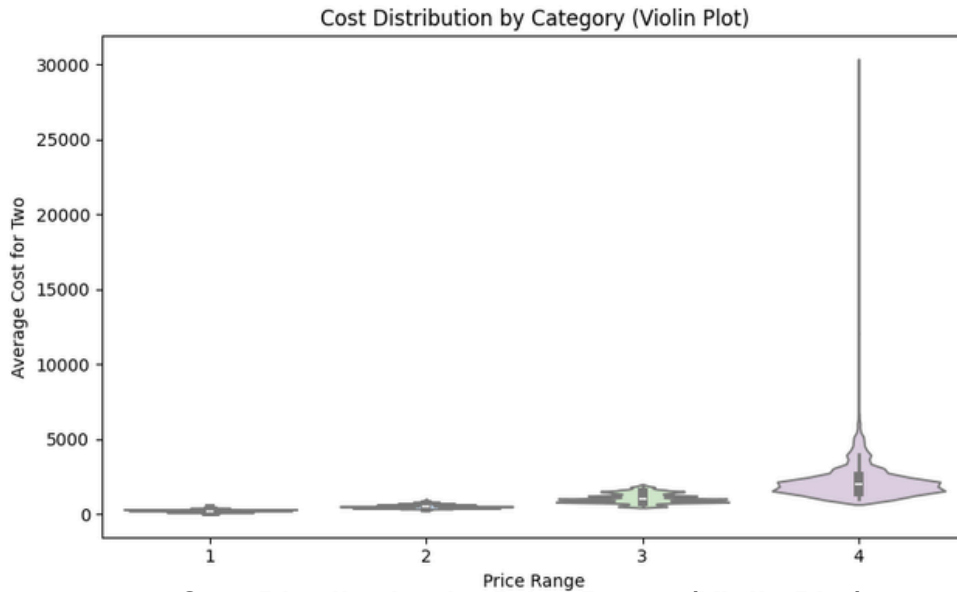
The strip plot shows how customer ratings vary across different price ranges. Across all four price categories, ratings are distributed fairly evenly between 2.0 and 5.0, indicating that both budget-friendly and premium restaurants can achieve high satisfaction. Price range 1 (low-cost restaurants) and price range 2 (mid-range) show the widest spread, with many restaurants rated above 4.0 but also a noticeable number with ratings below 3.0. Higher price ranges (3 and 4) generally maintain stronger clusters around 3.5 to 4.5, suggesting that premium restaurants are more consistent in quality, though they are not immune to low ratings. Interestingly, the presence of ratings at 0 across all ranges may represent outliers, non-rated entries, or poorly performing outlets. Overall, the plot highlights that customer satisfaction is not strictly tied to price—good and poor ratings exist across all categories—indicating that food quality and service matter more than pricing alone.



Average Cost for Two Across Price Categories

The lollipop chart highlights how the average cost for two increases across price categories. In Price Category 1, the average cost is the lowest (around ₹250), representing budget-friendly restaurants. Moving to Price Category 2, the average cost doubles to about ₹500, catering to mid-range dining. In Price Category 3, the cost rises significantly to around ₹1,100, reflecting premium dining experiences. Finally, Price Category 4 shows the steepest increase, with the average cost exceeding ₹2,200, representing high-end or luxury restaurants.

This clear upward trend demonstrates that the price categories are well-defined, with costs scaling consistently as one moves from budget to luxury dining. It also suggests that customers have a wide range of options depending on affordability, from low-cost casual eateries to exclusive fine-dining experiences.



Cost Distribution by Price Range (Violin Plot)

The violin plot provides a detailed look at how the average cost for two is spread across different price ranges:

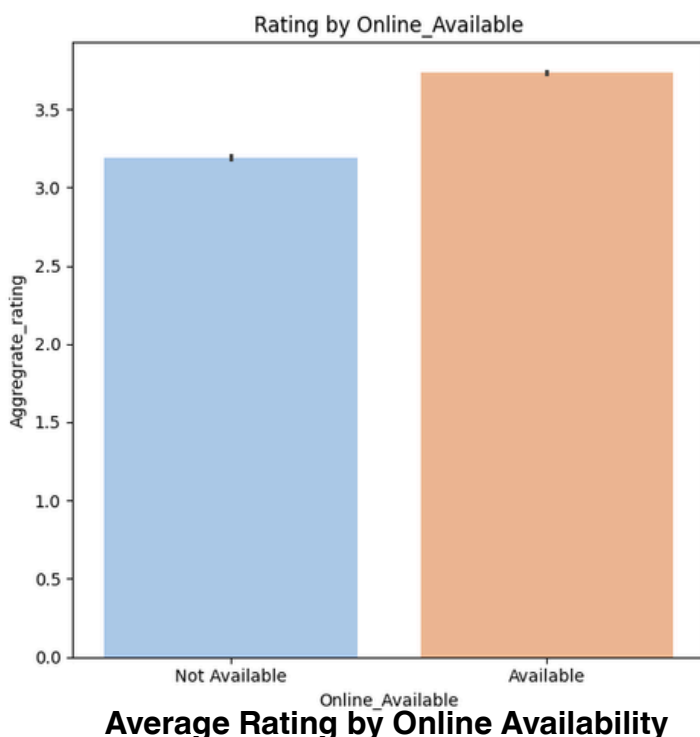
Price Range 1: Most restaurants fall within the ₹100–₹400 range, with very tight clustering, showing little variation.

Price Range 2: The distribution widens slightly, with costs ranging from ₹200 up to around ₹1,000. Still, the majority of restaurants stay near the ₹500 mark.

Price Range 3: A broader spread is visible, with costs typically ranging from ₹600–₹2,000, and a few outliers stretching even higher. This indicates a mix of mid-to-premium restaurants.

Price Range 4: The widest distribution by far, with costs starting around ₹1,000 but stretching to extreme outliers above ₹30,000. The density around ₹2,000–₹3,000 suggests most luxury restaurants cluster there, but the long tail shows a few ultra-premium establishments.

In summary, lower categories (1 & 2) have tighter and more predictable costs, while higher categories (3 & 4) show greater variation, with Price Range 4 including extreme luxury outliers.



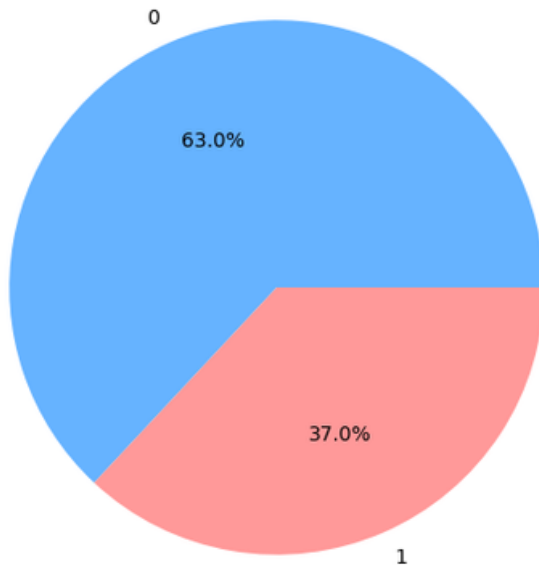
This bar chart highlights how online ordering availability impacts restaurant ratings:

Restaurants with Online Ordering Available
Average rating is ~3.7–3.8, noticeably higher. Suggests customers appreciate the convenience of online ordering and may associate it with better service.

Restaurants without Online Ordering
Average rating is lower, around 3.2. Indicates these outlets might be less preferred by customers in terms of accessibility and service flexibility.

Key Insight: Online availability has a positive correlation with higher ratings. Customers appear more satisfied with restaurants that provide online ordering options, likely due to convenience, reach, and better digital service integration.

Proportion of Restaurants Offering Table Booking



This pie chart shows the distribution of restaurants based on table booking availability:

63% of restaurants do not offer table booking – suggesting that the majority of dining establishments operate on a walk-in basis, which is common in casual dining and fast-food segments.

37% of restaurants offer table booking – these are more likely to be fine-dining or premium outlets, where reservations are necessary to manage customer flow and maintain service quality.

Key Insight: While most restaurants focus on quick and casual service, a significant portion (over one-third) recognizes the importance of reservations, especially in urban areas or higher-priced dining segments where customer planning and experience are crucial.

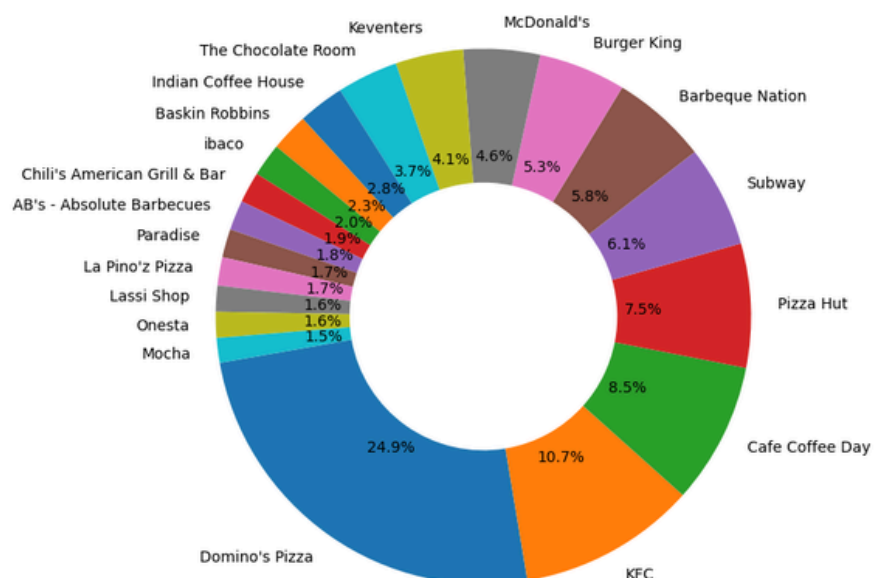
This pie chart shows the distribution of restaurants based on table booking availability:

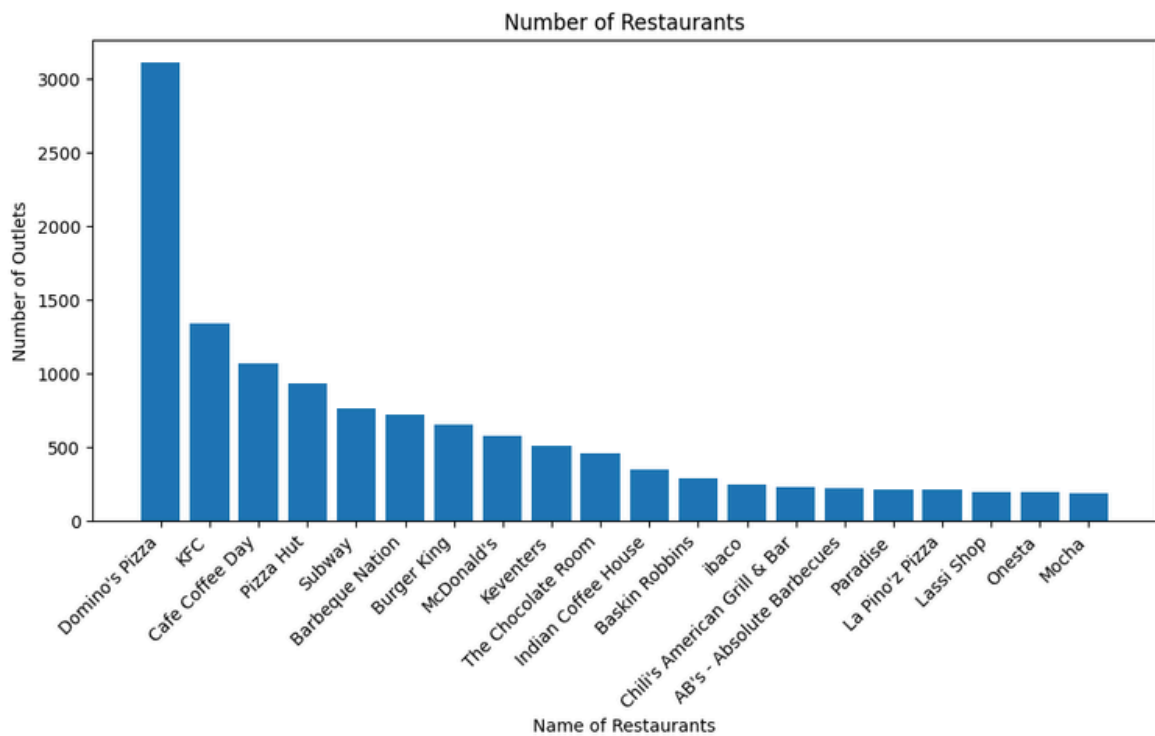
63% of restaurants do not offer table booking – suggesting that the majority of dining establishments operate on a walk-in basis, which is common in casual dining and fast-food segments.

37% of restaurants offer table booking – these are more likely to be fine-dining or premium outlets, where reservations are necessary to manage customer flow and maintain service quality.

Key Insight: While most restaurants focus on quick and casual service, a significant portion (over one-third) recognizes the importance of reservations, especially in urban areas or higher-priced dining segments where customer planning and experience are crucial.

Proportion of Restaurants Offering Table Booking





Number of Restaurants by Brand

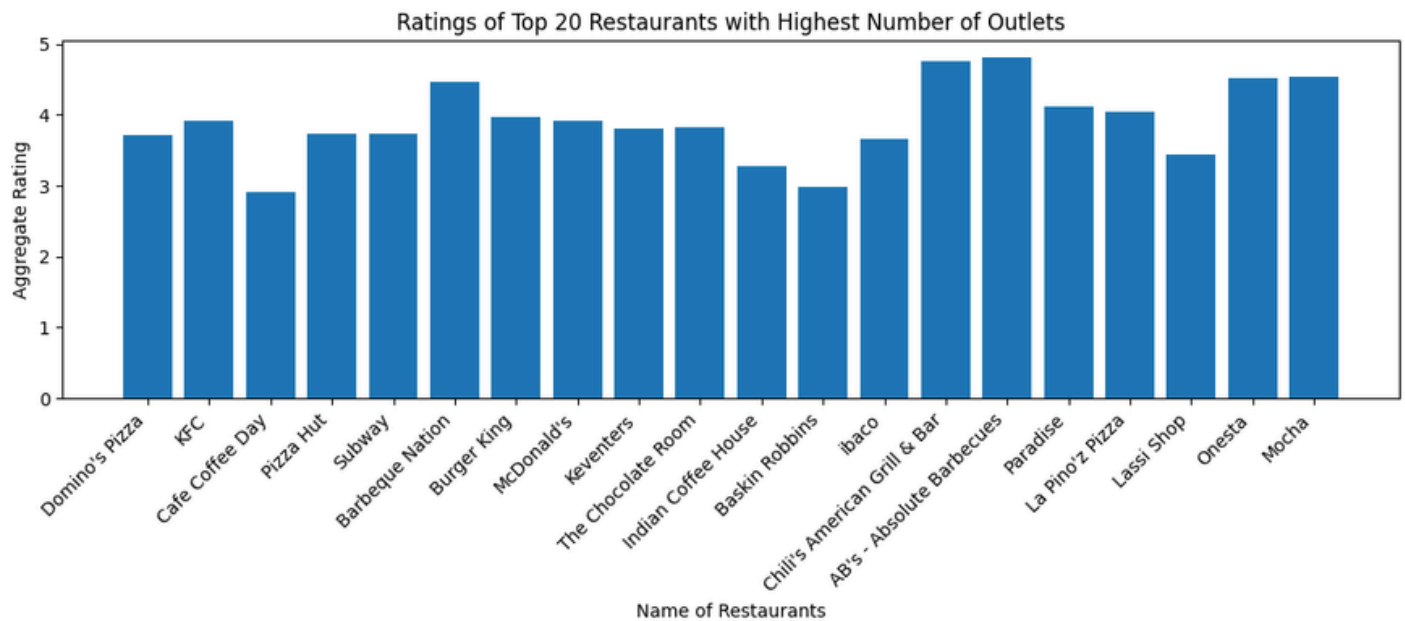
This bar chart displays the distribution of restaurant outlets by popular brands: Domino's Pizza leads by a large margin with over 3,000 outlets, reflecting its strong dominance in the quick-service restaurant (QSR) and pizza delivery segment. KFC and Café Coffee Day follow, with around 1,300 and 1,100 outlets respectively, showing their wide accessibility and consumer demand.

Other major chains like Pizza Hut, Subway, Barbeque Nation, Burger King, and McDonald's also have a strong presence with 600–900 outlets each, highlighting their consistent popularity across cities.

Mid-tier brands such as Baskin Robbins, The Chocolate Room, and Indian Coffee House have moderate representation, serving niche markets like desserts and coffee lovers.

Smaller but notable players like Paradise, La Pino'z Pizza, and Mocha have fewer outlets, yet remain regionally influential.

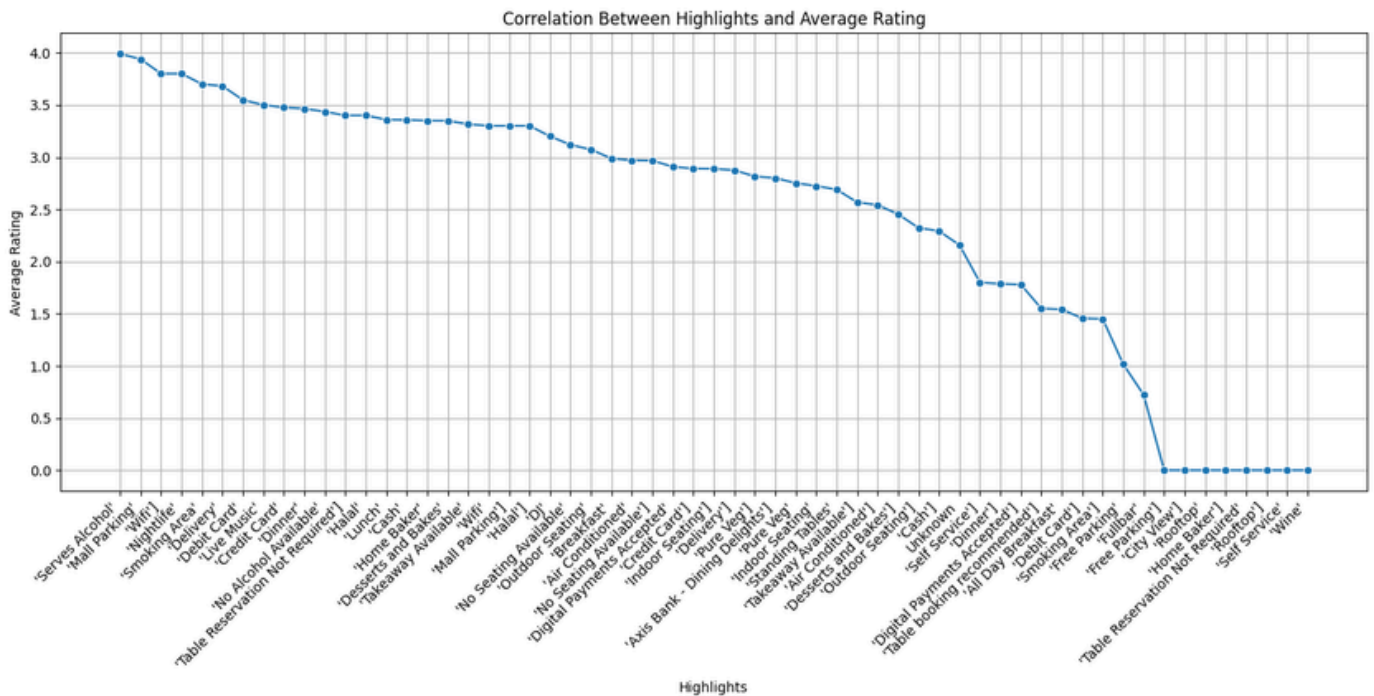
Key Insight: The restaurant industry is heavily dominated by fast-food and QSR giants like Domino's, KFC, and Pizza Hut, while local/regional brands maintain their presence by catering to specialized tastes or regional cuisines.



Ratings Analysis of Top 20 Popular Restaurant Chains

The chart above highlights the aggregate ratings of the top 20 restaurant chains with the highest number of outlets. Among them, AB's – Absolute Barbecues and Barbeque Nation stand out with the highest ratings, close to 4.8 and 4.5 respectively, reflecting strong customer satisfaction and popularity. Well-known global chains such as McDonald's, KFC, Domino's, and Pizza Hut receive average ratings ranging between 3.0 and 4.0, indicating consistent but moderate customer experiences. On the other hand, casual hangout places like Cafe Coffee Day and Baskin Robbins show relatively lower ratings, hovering around 3.0–3.5. Premium and niche restaurants such as Paradise, Onesta, and Mocha also perform well, maintaining ratings above 4.0.

Overall, the data suggests that specialized dining experiences and barbecue-style restaurants tend to achieve higher customer ratings compared to fast-food and café-style outlets, which often score closer to the average range.



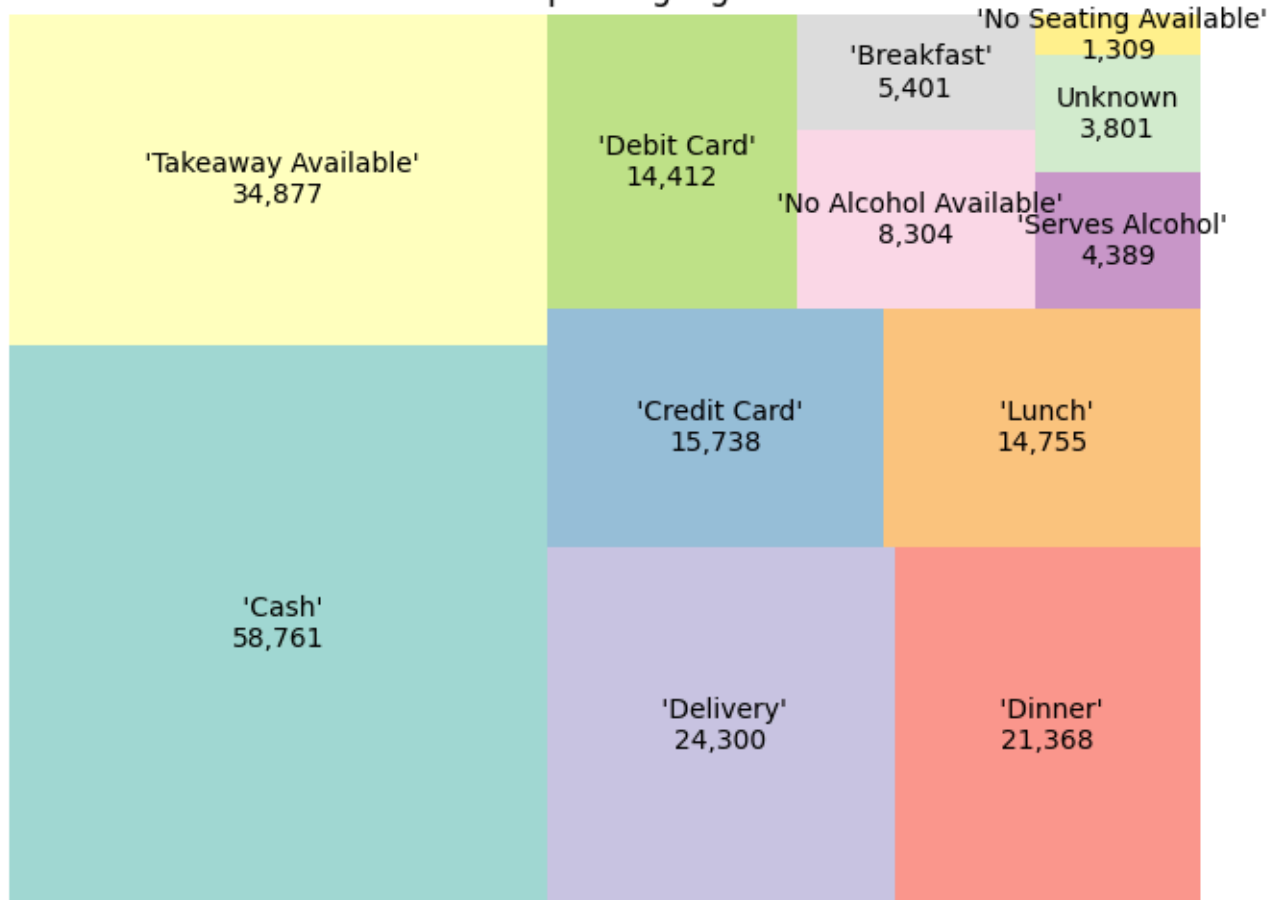
Correlation Between Restaurant Highlights and Average Ratings

The chart displays how different restaurant highlights (features/amenities) impact the average customer ratings. Outlets offering alcohol service, valet parking, nightlife, and smoking areas tend to receive the highest ratings (close to 4.0), indicating that such features strongly enhance customer satisfaction. Other positive contributors include live music, outdoor seating, home delivery, and buffet availability, which maintain ratings between 3.3 and 3.6.

On the other hand, features such as digital payments, indoor seating, and self-service fall in the middle range, with average ratings around 2.5–3.0, suggesting a more neutral impact. Interestingly, certain highlights like table booking not required, rooftop, city view, or self-service wine are associated with the lowest ratings (near 0.0–1.5), showing that they do not significantly improve customer experience, and in some cases, may even negatively influence it.

Overall, the data suggests that experience-driven amenities (alcohol, music, nightlife, valet parking) have the strongest positive correlation with ratings, while basic or niche services (self-service, rooftop, table booking not required) have minimal impact.

Treemap of Highlights



The treemap visualizes the most common highlights offered by restaurants, where the size of each block represents its frequency. The largest block, 'Cash' (58,761), indicates that cash payment is still the most widely accepted mode across restaurants. Other major highlights include 'Takeaway Available' (34,877), 'Delivery' (24,300), and 'Dinner' (21,368), reflecting that food convenience (takeaway/delivery) and dining options dominate customer preferences.

Medium-sized highlights such as 'Credit Card' (15,738), 'Lunch' (14,755), and 'Debit Card' (14,412) show the growing importance of digital and card-based transactions, along with flexible meal timings. Meanwhile, smaller but still relevant highlights include 'No Alcohol Available' (8,304), 'Breakfast' (5,401), and 'Serves Alcohol' (4,389), which highlight specific customer segments.

On the lower end, 'No Seating Available' (1,309) and 'Unknown' (3,801) appear as the least significant categories, likely tied to niche cases or incomplete data.

Overall, the treemap shows that basic conveniences (payments, delivery, takeaway, dining) dominate restaurant offerings, while alcohol and seating arrangements are less commonly highlighted.

Word Cloud of Review Categories



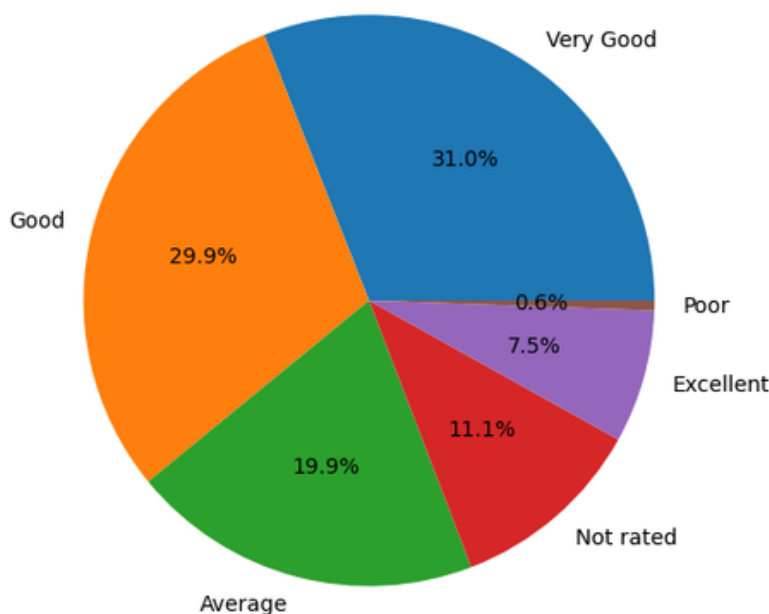
Word Cloud of Review Categories

The word cloud highlights the most frequent review sentiments given by customers. Larger words represent categories that appear more often in reviews. The most prominent categories are “Very Good” and “Good”, showing that a majority of customers have positive dining experiences. “Average” also appears significantly, suggesting that many customers find their experience satisfactory but not exceptional.

Smaller categories like “Excellent” represent top-rated experiences, while “Poor” is the least frequent, indicating that negative feedback is relatively rare compared to neutral or positive reviews.

Overall, the word cloud suggests that customer sentiment is skewed towards positive reviews, with most people rating restaurants as “Good” or “Very Good,” while extreme negative feedback remains minimal.

Percentage of Rating



The pie chart illustrates the proportion of ratings given by customers across different categories. The majority of reviews are positive: “Very Good” (31%) and “Good” (29.9%), together making up more than 60% of all ratings. This indicates that most customers are satisfied with their dining experiences.

“Average” ratings (19.9%) show that nearly one-fifth of customers found their experiences acceptable but not outstanding. A smaller share of reviews are “Excellent” (7.5%), representing highly satisfied customers.

On the other end, “Poor” ratings (0.6%) are negligible, suggesting that negative experiences are rare. Additionally, 11.1% of restaurants were “Not rated”, which could indicate either new restaurants or those with limited customer feedback.

Overall, the distribution confirms a strongly positive customer sentiment, with only a very small fraction of negative reviews.

Conclusion

The analysis of the restaurant dataset provides valuable insights into customer preferences, service features, and the overall dining landscape. It is evident that restaurants offering online ordering and table booking generally receive higher ratings, suggesting that convenience significantly influences customer satisfaction.

Additionally, popular chains such as Domino's Pizza, KFC, and Café Coffee Day dominate the market in terms of the number of outlets, reflecting strong brand presence and accessibility.

The highlights analysis revealed that services like takeaway, delivery, and cash payment are the most common features, while additional facilities such as serving alcohol or live music are offered by fewer restaurants but can lead to better ratings. The correlation study shows that factors such as parking availability, outdoor seating, and alcohol service positively impact customer ratings.

Review categories indicate that the majority of customers rate restaurants as Good or Very Good, with only a small fraction marking them as Poor. This demonstrates an overall positive dining experience in the dataset. Thus, the study concludes that customer satisfaction is highly dependent on service quality, accessibility, and value-added features, which restaurants should prioritize to remain competitive.

Acknowledgement

I would like to express my sincere gratitude to my faculty and mentors for their valuable guidance and constructive feedback during the course of this project. Their support has been instrumental in enhancing my understanding of data analysis, visualization, and interpretation techniques.

I am also thankful to the open-source community for providing access to powerful tools and libraries that made the implementation of this project possible. The availability of resources such as Python, Google Colab, and various visualization libraries greatly facilitated my work.

Finally, I take this opportunity to acknowledge my own dedication and consistent efforts in successfully completing this project. Working independently on this study has not only strengthened my technical skills but also improved my ability to carry out end-to-end research and analysis with discipline and responsibility.

Thank
you!

